Lecture 27

Power Series and Analytic Functions

All of the differential equations we have studied have solutions which can be expressed explicitly in terms of elementary functions (i.e. in terms of polynomials, exponentials, logarithms, trigonometric functions, etc.). Not all equations have explicit solutions and there are many important equations for which explicit solutions cannot be given or there is no known method to find one.

As we have seen in the previous lecture, we can use Taylor polynomials to approximate general solutions to differential equations. Not every function has a Taylor polynomial of all degrees at every point, however, for those functions which have such expansions, we can use a power series to express the solution to the equation regardless of whether the solution can be expressed explicitly.

We will begin by reviewing some properties of power series.

Definition: A **power series centered at x_0** is an expression of the form

$n=0,1,2,\ldots$

function of x $\longrightarrow$ $$\sum_{n=0}^{\infty} a_n(x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots$$

coefficient sequence

NOTE: We say a power series converges at $x = c$ if

$$\lim_{N\to\infty} \underbrace{\sum_{n=0}^{N} a_n(c-x_0)^n}_{\text{partial sum } S_N = \sum_{n=0}^{N} a_n(c-x_0)^n}$$

$x=c$

exists as a finite value. If this limit does not exist, we say the series diverges at $x = c$.

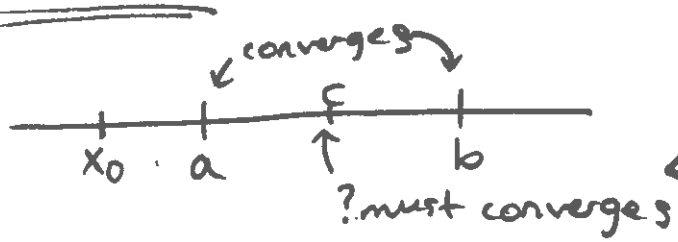
$x = x_0$

A power series converges wherever it is centered to a_0. Further, if a power series converges at two distinct values $x = a$ and $x = b$, then it also converges at any value *between* a and b.

$\sum a_n (a - x_0)^n$ ← converge

$\sum a_n (c - x_0)^n$ converge

$\sum a_n (b - x_0)^n$ converge



Theorem: There is a number ρ $(0 \le \rho \le \infty)$ called the **radius of convergence** of a power series such that the series converges absolutely for $|x - x_0| < \rho$ and diverges for $|x - x_0| > \rho$.

rho

If a series converges for all values of x, then $\rho = \infty$.

$x_0 - \rho$ $-\rho$ x_0 $+\rho$ $x_0 + \rho$

If it only converges at x_0, $\rho = 0$.

Theorem (Ratio Test)**:** If, for n large, the coefficients a_n are nonzero and satisfy

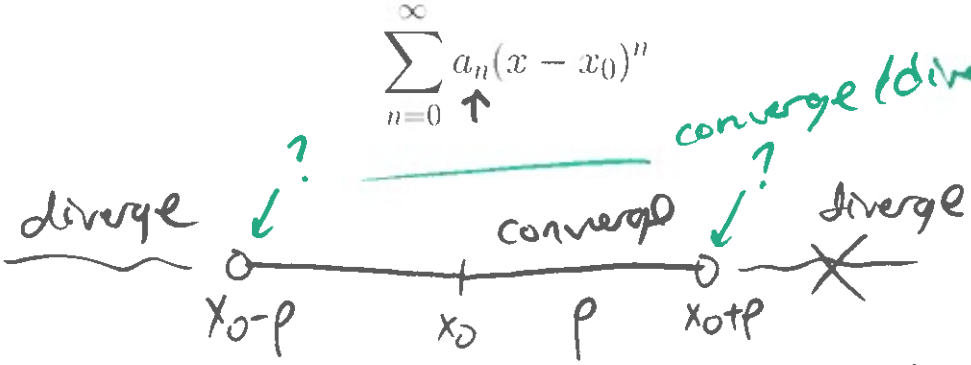
coefficients

$$\lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| = L$$

with $0 \le L < \infty$, then the radius of convergence of the power series

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

is $\rho = L$



$(x_0 - \rho, x_0 + \rho)$

endpoints $x_0 \pm \rho$ convergence must be calculated.

Example: Determine the radius of convergence and interval of convergence for the power series

$$\sum_{n=0}^{\infty} \underbrace{\frac{(n+2)!}{n!}}_{a_n}(x+2)^n \rightarrow a_n = \frac{(n+2)!}{n!}$$

$x - x_0$

$-2 = x_0$

Ratio Test: $\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{\frac{(n+2)!}{n!}}{\frac{(n+3)!}{(n+1)!}}\right|$

$= \lim_{n\to\infty}\left|\frac{(n+1)!\,(n+2)!}{n!\,(n+3)!}\right| = \lim_{n\to\infty}\left|\frac{(n+1)\cancel{n!}\,\cancel{(n+2)!}}{\cancel{n!}\,(n+3)\cancel{(n+2)!}}\right|$

$\boxed{\text{Recall: } n! = n(n-1)!}$

$= \lim_{n\to\infty}\left|\frac{n+1}{n+3}\right| = \lim_{n\to\infty}\frac{n+1}{n+3} = 1$

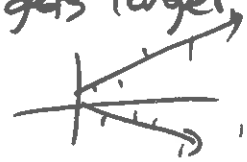
So $\boxed{\rho = 1.}$ center: -2, $\rho = 1$, $(-2-1, -2+1) = (-3, -1)$

We need to determine if we converge at $x = -3$ or $x = -1$.

$x = -3$: $\sum_{n=0}^{\infty}\frac{(n+2)!}{n!}(-3+2)^n = \sum_{n=0}^{\infty}\frac{(n+2)!}{n!}(-1)^n = \sum_{n=0}^{\infty}\frac{(n+2)(n+1)\cancel{n!}}{\cancel{n!}}(-1)^n$

diverges $\rightarrow$

$= \sum_{n=0}^{\infty}(n+2)(n+1)(-1)^n$, check $\lim_{n\to\infty}\underbrace{(n+2)(n+1)}_{\text{gets larger}}(-1)^n \neq 0$



$\boxed{\text{Divergence Test: } \sum_{n=0}^{\infty} b_n,\ \lim_{n\to\infty} b_n \neq 0, \text{ diverges}}$

$x = -1$: $\sum_{n=0}^{\infty}(n+2)(n+1)(-1+2)^n = \sum_{n=0}^{\infty}(n+2)(n+1)$ $\lim_{n\to\infty}(n+2)(n+1) \neq 0$

diverges.

IOC: $\boxed{(-3, -1)}$

Example: Determine the radius of convergence and interval of convergence for the power seies

$$\sum_{n=0}^{\infty} \boxed{\frac{(-2)^n}{n+1}}(x-3)^n \qquad a_n = \frac{(-2)^n}{n+1}$$

center: 3

Ratio Test: $\lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{\frac{(-2)^n}{n+1}}{\frac{(-2)^{n+1}}{n+2}}\right|$

$$= \lim_{n\to\infty}\left|\frac{\cancel{(-2)^n}(n+2)}{(-2)^{\cancel{n}+1}(n+1)}\right| = \lim_{n\to\infty}\left|\frac{n+2}{(-2)(n+1)}\right|$$

$$= \frac{1}{2}\underbrace{\lim_{n\to\infty}\frac{n+2}{n+1}}_{1} = \frac{1}{2} = \rho. \text{ So } |x-3| < \frac{1}{2}$$

$$\Rightarrow \left(3-\frac{1}{2}, 3+\frac{1}{2}\right) = \left(\frac{5}{2}, \frac{7}{2}\right)$$

Check endpoints: $x = \frac{5}{2}$: $\sum_{n=0}^{\infty}\frac{(-2)^n}{n+1}\left(\frac{5}{2}-3\right)^n = \sum_{n=0}^{\infty}\frac{\cancel{(-2)^n}}{n+1}\cancel{\left(-\frac{1}{2}\right)^n}$

$$= \sum_{n=0}^{\infty}\frac{1}{n+1} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$$

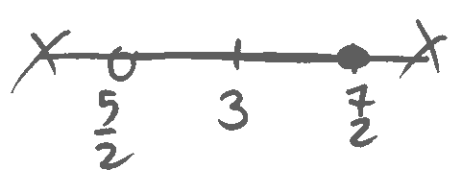
harmonic series <u>diverges</u>

$x = \frac{7}{2}$: $\sum_{n=0}^{\infty}\frac{(-2)^n}{n+1}\left(\frac{1}{2}\right)^n = \sum_{n=0}^{\infty}\frac{(-1)^n\cancel{2^n}}{n+1}\cdot\cancel{\frac{1}{2^n}}$

$$= \sum_{n=0}^{\infty}\frac{(-1)^n}{n+1} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

alternating harmonic series <u>converges</u>.

So IOC: $\left(\frac{5}{2}, \frac{7}{2}\right]$



There will be situations where we need to alter the indices in our summation notation in order to be able to combine various summations together.

$$\sum_{n=1}^{\infty}(n+1)x^{n-1} = \sum_{n=0}^{\infty}\left((n+1)+1\right)x^{(n+1)-1}$$

$\Rightarrow n = u+1$

$u = n-1$

$n=1,\ u=0$

down 1 $\downarrow$ $n=0$

$\uparrow$ up 1

$$= \sum_{n=0}^{\infty}(n+2)x^{n}$$

$$\sum_{n=1}^{\infty}\frac{n}{n+3}x^{n+3} = \sum_{n=4}^{\infty}\frac{(n-3)}{(n-3)+3}x^{(n-3)+3}$$

up 3 $\uparrow$ $\downarrow$ down 3

$$= \sum_{n=4}^{\infty}\frac{n-3}{n}x^{n}$$

We need both index and exponent to align in order to combine series together.

Example: Write $\sum_{n=1}^{\infty} 2na_nx^{n-1} + \sum_{n=0}^{\infty} 6a_nx^{n+1}$ as one series.

(1) Need exponents to align (x^n)
(2) If indices align (good), otherwise remove terms until they align
(3) Combine

$$= \sum_{n=0}^{\infty} 2(n+1)a_{n+1}x^{n} + \sum_{n=1}^{\infty} 6a_{n-1}x^{n}$$

separate $n=0$ term

$$= \Big[\underbrace{2(0+1)a_{0+1}x^{0}}_{n=0} + \underbrace{\sum_{n=1}^{\infty} 2(n+1)a_{n+1}x^{n}}_{\text{rest}}\Big] + \sum_{n=1}^{\infty} 6a_{n-1}x^{n}$$

$$= 2a_1 + \sum_{n=1}^{\infty}\left[2(n+1)a_{n+1} + 6a_{n-1}\right]x^{n}$$

Theorem: If $\sum_{n=0}^{\infty} a_n(x-x_0)^n = 0$ for all x in some interval, then $a_n = 0$ for all n.

$= 0$

Example: Find an expression for a_n as a function of n if

$$\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=-1}^{\infty} a_{n+1} x^{n+1} = 0.$$

Does this combination of series equal an explicit function?

$$\Rightarrow \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \Big[\underbrace{(n+1) a_{n+1} + a_n}_{=0} \Big] x^n = 0$$

$$\Rightarrow (n+1) a_{n+1} + a_n = 0 \quad \text{for all } n = 0, 1, 2, \ldots$$

Solve for highest indexed coefficient:

$$(n+1) a_{n+1} = -a_n \Rightarrow \boxed{a_{n+1} = -\frac{a_n}{n+1}}$$

recurrence relation

key

$$n=0: \quad a_1 = -\frac{a_0}{0+1} = -a_0 = -\frac{a_0}{1!}$$

$$n=1: \quad a_2 = -\frac{a_1}{1+1} = -\frac{1}{2} a_1 = -\frac{1}{2}(-a_0) = \frac{a_0}{2} = \frac{a_0}{2!}$$

$$n=2: \quad a_3 = -\frac{a_2}{2+1} = -\frac{1}{3} a_2 = -\frac{1}{3} \cdot \frac{a_0}{2} = -\frac{a_0}{6} = -\frac{a_0}{3!}$$

pattern

$$n=3: \quad a_4 = -\frac{a_3}{3+1} = -\frac{1}{4} a_3 = -\frac{1}{4}\left(-\frac{a_0}{6}\right) = \frac{a_0}{24} = \frac{a_0}{4!}$$

So $a_i = (-1)^{i+1} \dfrac{a_0}{(i+1)!}$ or $\boxed{a_n = \dfrac{(-1)^{n+1} a_0}{(n+1)!}}$, $n \geq 0$.

if $a_n = \dfrac{(-1)^{n+1} a_0}{(n+1)!}$ then $\displaystyle\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(n+1)!} a_0 x^n = (-e^{-x} + 1) a_0$

constant

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$$

Theorem: If $f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n$ has a positive radius of convergence ρ, then for $|x-x_0| < \rho$, $f(x)$ is both differentiable and integrable and we have $\rho > 0$

$(x_0-\rho, x_0+\rho)$

- $f'(x) = \sum_{n=1}^{\infty} na_n(x-x_0)^{n-1}$

for $|x-x_0|<\rho$

- $\int f(x)\,dx = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-x_0)^{n+1}$

Example: Recall that $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ ← geometric series with radius of convergence $\rho = 1$ and interval of convergence $(-1,1)$. Use this to find a power series expansion for the following functions as well as their intervals of convergence.

- $f(x) = \ln(1-x)$ $\Rightarrow f'(x) = \frac{1}{1-x}(-1) = \frac{-1}{1-x} = -\sum_{n=0}^{\infty} x^n$

$$\ln(1-x) = \int f'(x)dx = \int -\sum_{n=0}^{\infty} x^n\,dx = -\sum_{n=0}^{\infty} \frac{1}{n+1}x^{n+1} + C$$

!!

$$x=0,\quad \ln(1-0) = \ln(1) = 0 = -\cancel{\sum_{n=0}^{\infty} \frac{1}{n+1}(0)^{n+1}} + C$$

$$= C \Rightarrow \boxed{C=0}$$

So $\boxed{\ln(1-x) = -\sum_{n=0}^{\infty} \frac{1}{n+1}x^{n+1}}$ for $(-1,1)$

$x=1$, $-\sum_{n=0}^{\infty} \frac{1}{n+1} \to \infty$ harmonic series, $x=-1$: $-\sum_{n=0}^{\infty} \frac{1}{n+1}(-1)^{n+1}$ converges.

IOC: $[-1,1)$.

- $f(x) = \dfrac{1}{(1+x)^2}$, $\displaystyle\int \frac{1}{(1+x)^2}\,dx = \int (1+x)^{-2}\,dx = -(1+x)^{-1} + C$

Thus $\dfrac{d}{dx}\left(\dfrac{1}{1+x}\right) = \dfrac{-1}{(1+x)^2} = -f(x)$

$$\boxed{\frac{1}{1+x} = \sum_{n=0}^{\infty}(-x)^n = \sum_{n=0}^{\infty}(-1)^n x^n}$$

$$\boxed{\frac{1}{1-(-x)}}$$

$$\Rightarrow f(x) = -\frac{d}{dx}\left(\sum_{n=0}^{\infty}(-1)^n x^n\right)$$

$$= -\sum_{n=1}^{\infty}(-1)^n n x^{n-1} = \sum_{n=1}^{\infty}(-1)^{n+1} n x^{n-1}$$

For $(-1,1)$

diverge for $x = -1, 1$.

$$\boxed{f(x) = \sum_{n=1}^{\infty}(-1)^{n+1} n x^{n-1}}$$

$$\boxed{\text{IOC}: (-1,1).}$$

- $f(x) = \arctan(x)$ $\qquad \dfrac{d}{dx}\arctan(x) = \dfrac{1}{1+x^2}$

$$f'(x) = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty}(-x^2)^n = \sum_{n=0}^{\infty}(-1)^n x^{2n}$$

So $\arctan(x) = f(x) = \displaystyle\int f'(x)\,dx = \int \sum_{n=0}^{\infty}(-1)^n x^{2n}\,dx$

$$= \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{2n+1} + C$$

$$0 = \arctan(0) = \cancel{\sum_{n=0}^{\infty}(-1)^n \frac{0^{2n+1}}{2n+1}} + C \Rightarrow C = 0.$$

$(-1,1)$

$$\boxed{f(x) = \arctan(x) = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{2n+1}}$$

$$\boxed{\text{IOC}: (-1,1]}$$

Definition: A function $f(x)$ is **analytic** at x_0 if it can be expressed as

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ $\rho = \infty$

analytic at $x=0$.

with a positive radius of convergence.

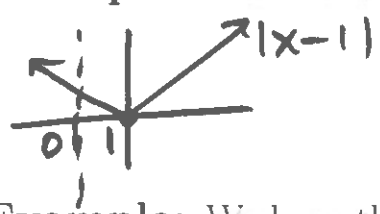
$\uparrow$

$\rho > 0$

$\sin(x)$, $\cos(x)$ analytic at $x=0$.

$\ln(1-x)$ analytic at $x=0$.

Example: We have that $f(x) = |x-1|$ is not analytic at $x = 1$.



$f'(x)$ DNE at $x=1$.

Example: We have that $f(x) = x^{7/3}$ is not analytic at $x = 0$.

$f'(x) = \frac{7}{3}x^{4/3}$, $f''(x) = \frac{28}{9}x^{1/3}$, $f'''(x) = \frac{28}{27}x^{-1/3}$

$\uparrow$

DNE at $x=0$.

At a specific center, any power series representation of $f(x)$ is the same.

All in all, if a function $f(x)$ is analytic at $x = x_0$, then we can express the function $f(x)$ using its Taylor series centered at x_0 and write

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

It will follow that we will want to find where our differential equations have analytic solutions so we can write them as a power series but not necessarily explicitly in terms of elementary functions.